# Renewal → Scheduling Pipeline: Audit, Re-Engineering & Rollout

Bambinos.live · AI Program Manager (Operations) take-home · Deliverables A, B, D (C = repo prototype) · Data snapshot read as of 4-Nov-2026 09:00 IST

> **Headline.** In the 13-invoice sample, **11 of 12 parents who paid have no scheduled class** (91.4% of cash; ₹126,850 of ₹138,850). Teachers are not the main bottleneck: they file the form in a median **27 min**. The pipeline stalls after that: **7 of 11 forms sit unapproved for over 12 h**, and 3 of the 4 approved ones die in the last step. The engine in the repo would auto-schedule **5 of 13** now, and about **8 of 12 paid** once orphan pre-fill and payer confirmation are live. The other cases go to one named owner each, with a drafted message and an SLA.

## A. Root-cause problem definition & leakage audit

### A1. Where the 4-table chain leaks (sample data)

| **invoices** | **→ form** | **→ booking_request** | **→ class_schedule** |
|---|---|---|---|
| 13 rows · 12 paid/partial · 1 failed | 2 paid invoices never get a form (orphans)<br>1 form on a failed invoice<br>1 duplicate, 1 wrong invoice ID | 7 of 11 forms unapproved >12 h<br>only 4 approved | 3 of 4 BRs stuck<br>1 scheduled, with 2 of 24 sessions |

| Invoice | Payment | Booking req. | Sessions sched. | Engine decision | Reason codes |
|---|---|---|---|---|---|
| 84901 | Paid | Completed | 2/24 | SCHEDULED BUT FLAGGED | SCHEDULE_INCOMPLETE |
| 84902 | Paid | Pending_Batch_Allocation | 0/24 | AUTO APPROVE AUTOFIX | STUCK_PENDING_ALLOCATION |
| 84903 | Paid | | 0/24 | AUTO APPROVE | FORM_PENDING_AGEING |
| 84904 | Paid | | 0/24 | HOLD EXCEPTION | PAYER_IDENTITY_MISMATCH |
| 84905 | Paid | Failed_Batch_Full | 0/24 | AUTO APPROVE AUTOFIX | FALSE_BATCH_FULL |
| 84906 | Paid | | 0/ | HOLD EXCEPTION | ORPHANED_INVOICE |
| 84907 | Paid | | 0/24 | HOLD EXCEPTION | DISCOUNT_NOT_AUTHORISED |
| 84908 | Paid | | 0/24 | AUTO APPROVE AUTOFIX | DUPLICATE_FORM FORM_PENDING_AGEING |
| 84909 | Partially_Paid | | 0/24 | HOLD EXCEPTION | PARTIAL_PAYMENT AMOUNT_MISMATCH |
| 84910 | Paid | | 0/ | HOLD EXCEPTION | ORPHANED_INVOICE |
| 84911 | Failed | | 0/24 | HOLD EXCEPTION | PAYMENT_NOT_CONFIRMED |

| | | | | | |
|---|---|---|---|---|---|
| 84912 | Paid | Awaiting_Parent_Batch_Selection | 0/24 | HOLD EXCEPTION | BATCH_CHANGE_NO_MATCH |
| 84913 | Paid | | 0/24 | AUTO APPROVE AUTOFIX | INVOICE_ID_TYPO FORM_PENDING_AGEING |

## A2. Hidden failure modes found

| # | Failure mode | Evidence in data | Why it happens |
|---|---|---|---|
| 1 | **Orphaned invoice** (paid, no form) | 84906 (GCC, ₹18,000, 41 h), 84910 (₹11,000, 22 h) | Step 2 depends on a teacher remembering. Nothing watches for the missing row. |
| 2 | **Ops approval queue** is the real bottleneck | 7 of 11 forms pending, oldest 48 h (84903, a clean case). Median 1.5 h when Ops does act. | The check is manual copy-compare. There is no queue, SLA or weekend cover. 1 and 2 Nov were Sun/Mon. |
| 3 | **Approved ≠ scheduled** | BR 5502 pending allocation 64 h; 5505 "Failed_Batch_Full"; 5512 awaiting parent | Step 4 has no owner and no alert. Failures sit silently in a status column. |
| 4 | **False "batch full"** | 84905: batch 9205 is 6/6, and the student's own seat is one of the 6 | The capacity check counts the renewing child as a competitor for their own seat. |
| 5 | **Batch-change renewal with no slot** | 84912: parent wants Sat/Sun 11:00 for Gita L1, but the only Gita L1 batch is Wed/Fri 16:30. BR has no batch and has been idle 16 h | Nobody checks availability before the parent is promised a slot. Same-batch and batch-change use the same path. |
| 6 | **Payer ≠ registered parent** | 84904: invoice mobile ...77 and a different email, form says "Mother paid from her number" | Verification is by eyeball. A real mismatch looks the same as a fraud or wrong-student case. |
| 7 | **Off-book discount** | 84907: 20% "retention scholarship", ₹7,600 vs ₹9,500, in the remarks only | Teachers can offer discounts in free text. No approval trail or code. |
| 8 | **Partial payment treated as full** | 84909: instalment 1 of 2 (₹5,250 of ₹10,500). Teacher entered ₹10,500 and 24 sessions. | The form has no payment-type field. The teacher types whatever they believe. |
| 9 | **Unpaid invoice with a "proof" screenshot** | 84911: Failed, no gateway payment ID, form attached with a UPI "processing" screenshot | Teachers trust parents' screenshots. Only Finance can confirm bank capture. |
| 10 | **Typos and duplicates** | 3313 cites invoice 84931 (should be 84913) and the link has the same typo. 3308 and 3309 are the same invoice 5 min apart. | Free-text invoice link and ID, with no foreign-key validation. |
| 11 | **Partial session generation** | 84901: 24 purchased, 2 bookings exist, so 22 sessions (92%) are missing. The paid-for sessions run out after 5-Nov. | Nothing compares sessions purchased to sessions scheduled. |
| 12 | **Class-gap risk** | 84902, 84903, 84910 (an orphan) and 84912 (waiting on a batch change) have a batch class **today** (Wed) with no booking. 84913 has one tomorrow. | Time-to-next-class is not part of any priority rule. |

Structural observation: `booking_requests` is a near-copy of an approved form, and `faculty_form_unapproved` re-keys data the invoice already holds (student, amount, link). Three tables with human hand-offs produce the failures above.

## A3. Metrics and SLAs that are broken or invisible

**11 / 12**
paid, not scheduled (all >12 h)

**16.7%**
orphaned invoice rate (2 of 12 paid)

**64%**
forms pending >12 h (7/11)

**75%**
approved but not scheduled (3/4)

| Metric | Definition | Target SLA | Today (sample) |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **Payment-to-Schedule turnaround** | invoice.paid_at → first class_schedule_bookings.created_at | Happy path ≤ 15 min; P90 ≤ 4 h; hard cap 24 h | 1 of 12 scheduled (1.7 h). Other 11 are 16-67 h and still open. |
| Orphaned Invoice Rate | paid invoices with no form after 2 h ÷ paid invoices | < 2%, each orphan resolved within 4 h | 16.7% |
| Form approval latency | form.created_at → approved_at (or auto) | Auto: < 1 min. Manual: ≤ 2 h. | Median 1.5 h when done; 7 of 11 never done in 12 h+ |
| Approved-not-scheduled rate | BRs without class_schedule_bookings after 4 h | 0% | 75% (3 of 4) |
| Schedule completeness | sessions scheduled ÷ sessions purchased | 100% (or rolling 4-week horizon) | 2 of 24 (8%) on the only scheduled case |
| First-pass auto-approval rate | renewals scheduled with no human touch | ≥ 65% by day 90 | 0% today. The engine gets 38% of invoices now. |
| First-class continuity rate | paid renewals scheduled before the next batch class ÷ paid renewals | > 98% | 1 of 12 (8%) |
| Batch overflow rate | booking requests blocked by a capacity check ÷ booking requests | 0% (seat lock at checkout) | 25% (1 of 4, and arguably a false block) |
| Class-gap exposure | paid students whose next batch class starts within 48 h with no booking | 0 | 5 (84902, 84903, 84910, 84912 today; 84913 tomorrow) |
| Exception ageing | % of flagged items open past reason-code SLA | < 10% | 7 of 8 held invoices already past SLA |
| Data-integrity rate | forms with dup/typo/ID mismatch ÷ forms | < 1% | 2 of 12 (17%) |
| Revenue-at-risk | ₹ paid and unscheduled >12 h | ₹0 | ₹126,850 |

Caveat: 13 invoices is a small sample from one window. The rates show where the process fails, not what the population rates are. Section D includes a plan to re-baseline them on the full table.

# B. Re-engineered "Zero-Touch" workflow

## B1. Why each manual step exists, and whether it should

| Step | Why it exists | Verdict |
|---|---|---|
| 2. Teacher fills the form | The invoice doesn't know the parent's intent (same batch? new slot?) or the teacher's offer | **Eliminate for happy path.** Embed `student_id`, `classbatch_id` and `educator_id` in the Razorpay Notes / Stripe metadata, so the webhook carries verified context. Capture intent at payment-link creation: the teacher (or the parent on the success page) taps "Continue in same batch" or "Change batch". The invoice then carries student, batch, teacher and sessions. The teacher is only pinged for exceptions. |
| 3. Ops verifies the invoice | Trust: typos, wrong amounts, unpaid links | **Replace with a machine check.** The gateway webhook proves payment. Foreign keys prove identity. A rules engine compares amount, batch, contact and capacity. Humans only see mismatches. |
| 4. Parent/Ops picks batch | Needed only if the student is changing batch | **Skip for same-batch.** The seat is already held. For batch change, offer only batches with real seats, on WhatsApp, with a 24 h expiry. |

## B2. Target flow (n8n / Make, Postgres or Sheets, WhatsApp Business API, Slack)

**1 Trigger**
Razorpay/Stripe `payment.captured` webhook, or invoice insert

**2 Enrich**
Lookup student, batch, seat count, fee from reference table

**3 Rules**
7-point check (policy file). Own seat excluded from capacity.

**4a Happy**
Create BR, create all 24 sessions, WhatsApp parent and teacher

**4b Exception**
LLM triage, route to one owner, draft message, log to queue

**5 Sweeper (every 30 min, 7 days)**
Orphans, forms >2 h, BRs idle >1 h, sessions < purchased, next-class < 48 h

**6 Escalation ladder**
T+2 h owner · T+4 h Ops L1 · T+12 h Ops Head · class < 24 h → page on-call

**7 Dashboard**
Metabase on the audit table. Daily 9 am Slack digest.

**Happy path that skips all manual steps:** paid in full, same batch, same teacher, registered contact, standard fee, seat held. In the sample this is 84902, 84903, 84905, 84908 and 84913 (84913 and 84908 after auto-fix). Parent pays, a WhatsApp confirmation with the class link arrives within minutes, and the teacher gets a read-only FYI. **Teacher and Ops touches: 0.**

### B3. Smart exception handling (triage → owner → message)

| Mismatch | Reason code | Auto action | One owner | SLA |
|---|---|---|---|---|
| Paid, no form | ORPHANED_INVOICE | WhatsApp to teacher: "1 = same batch, 2 = change". One reply creates the form. Parent gets a "we're on it" note. | Teacher → Ops L1 at 4 h | 4 h |
| Payer mobile/email ≠ registered | PAYER_IDENTITY_MISMATCH | YES-confirm WhatsApp to the *registered* number. YES triggers auto-approve. | Parent → Ops L1 call at 4 h | 8 h |
| Form amount ≠ invoice | AMOUNT_MISMATCH | Show the teacher both numbers with a one-tap "use invoice amount". | Teacher | 8 h |
| Instalment payment | PARTIAL_PAYMENT | Schedule sessions covered (policy: first 12). Auto-remind the parent for instalment 2. | Finance | 8 h |
| Net < standard fee | DISCOUNT_NOT_AUTHORISED | Approval card to the Ops Head. Teacher is asked for the scholarship code. | Ops Head | 8 h |
| Failed payment + screenshot | PAYMENT_NOT_CONFIRMED | Pull the gateway status. Teacher is told not to promise slots. Parent is asked for the UTR. | Finance | 4 h |
| Batch change, no matching slot | BATCH_CHANGE_NO_MATCH | Offer the parent real alternatives. At 3+ requests for the same slot, suggest opening a new batch. | Ops Scheduler | 12 h |
| Typo, duplicate, false "full", stuck BR | INVOICE_ID_TYPO · DUPLICATE_FORM · FALSE_BATCH_FULL · STUCK_PENDING_ALLOCATION | **Auto-healed and logged.** No human is involved. | — | — |

Design principle: rules decide and are auditable. The LLM reads teacher remarks, resolves ambiguity, and writes messages. If confidence is below 0.8 the case is held for a human. The LLM never approves money-related cases. The prompt is in `prompts/triage_agent.md`. Remarks only change who is contacted first, never whether a blocking flag applies.

### B4. Prototype results (Deliverable C, run on the provided CSVs)

**5**
auto-approved for class_schedule_bookings (38%)

**8**
invoices flagged with reason codes

**18**
drafted WhatsApp/Slack messages

**4**
silent failures auto-healed

Outputs: `output/1_auto_approved…csv`, `2_flagged_exceptions.csv`, `3_drafted_messages.md`. The engine detects 10 of the 12 failure modes in A2 (class-gap exposure and the structural issues are dashboard and design items). Today's sample has 5 auto-approvals, 3 forms that can be cleared on a

single parent or teacher reply (84904, 84906, 84910), and 5 that need real decisions (discount, instalment, failed payment, no slot, short schedule).

## D. Ops Head dashboard, SOP & 30-60-90 plan

### D1. Morning dashboard (Metabase on `pipeline_audit`, mock with today's sample numbers)

| | | | |
|---|---|---|---|
| **11** Paid invoices unscheduled >12 h | **7** Forms awaiting approval >12 h | **3** Approved BRs not scheduled | **5** Students with a class <48 h and no booking |
| **₹126,850** Revenue paid, unscheduled | **38%** First-pass auto-approval | **2** Orphaned invoices | **~44 h/wk** Ops hours saved at 300 renewals/wk (est.) |

| Panel | Content | Action it drives |
|---|---|---|
| 1. Red list | Every paid invoice unscheduled >12 h, sorted by hours to next class. Shows reason code, owner, SLA clock. | Ops Head sees the day's fires in 30 seconds |
| 2. Funnel | Paid → form → approved → BR → scheduled, with median hours between stages | Shows which stage leaks (today: approval and allocation) |
| 3. Exceptions by owner | Open items per teacher/Finance/Ops L1, with breach % | Rebalance load, coach slow teachers |
| 4. Teacher scorecard | Orphan rate, form errors, discount offers per educator | Targeted coaching |
| 5. Auto-rate & hours saved | Weekly trend vs 65% target; hours saved = auto-approved × 6 min + chases avoided × 15 min | Proves ROI; unlocks the next rollout stage |
| 6. Capacity & demand | Batch-change requests with no slot, grouped by class/slot | Decide on new batches |

**Hours-saved assumptions (to be replaced by a one-week time study in Phase 0):** 300 renewals/week; today 6 min manual verification each (30 h) plus 15 min of chasing on about 40% of cases (30 h) = ~60 h/week. Target: 65% auto, 35% exceptions at about 4 min triage plus 5 min of chasing with drafted messages = ~16 h/week, saving ~44 h/week (about 70%) before counting recovered revenue and fewer parent complaints.

### D2. SOP for Ops and teachers (what changes on Monday)

| Who | Before | After |
|---|---|---|
| Teacher (200+) | Fill a form after every payment | Pitch → tap "same batch" in the payment link flow. Answer a WhatsApp "1/2" only when pinged. No new app. |
| Ops L1 (10 people) | Check every row, chase manually | Work only the exception queue in order of SLA clock. Use drafted messages: Send / Edit / Reassign. One owner per item. |
| Ops Scheduler | Allocate batches ad hoc | Handles BATCH_CHANGE and SCHEDULE_INCOMPLETE only, with real seat options shown |
| Finance | Called ad hoc | Owns PAYMENT_NOT_CONFIRMED and PARTIAL_PAYMENT, with a 4 h / 8 h SLA |
| Ops Head | No visibility | 9 am dashboard review. Approves discounts. Weekly review of SLA breaches and auto-rate. |

### D3. Rollout without breaking live classes (change management)

| Phase | Scope | Guardrails and exit gate |
|---|---|---|

| | | |
|---|---|---|
| **Days 0-30: Shadow** | Engine runs read-only on live data beside the manual process. Ops compares every decision. Re-baseline all metrics on the full invoice table. Time-study the manual work. Get WhatsApp templates approved. Fix the capacity check. Add invoice-ID validation and the "same batch / change" option at link creation. | No writes to production. Gate: ≥ 98% agreement with Ops, zero false approvals across ≥ 500 historical renewals, ≥ 60% zero-touch yield, 100% detection of class gaps < 48 h, sign-off from the Ops Head and the engineering lead. Also patch the scheduler that created only 2 of 24 sessions, and confirm the seat-count rule with engineering. |
| **Days 31-60: Pilot** | Auto-approve for 20 teachers in 1-2 categories, same-batch full payments only. Auto-nudges for orphans and ageing forms. Ops reviews a daily sample of 10% of auto-approvals. | Kill switch to fall back to manual in one click. Anything with a class within 24 h gets a human double-check. Exception ladder: L1 Ops (identity, forms, 6-8 h) · L2 Scheduling/Academic Ops (orphans with a class < 48 h, batch changes, 2-4 h) · L3 Finance/Ops Head (discounts, instalments, failed payments, 4 h). Gate: 0 wrong schedules, orphan rate < 5%, median pay-to-schedule < 1 h. |
| **Days 61-90: Scale** | All 200+ teachers in cohorts of ~40 per week. Retire the form for happy-path cases. Batch-change, identity mismatch and instalments go live. Handover SOP, dashboards, on-call rota with weekend cover. | Training: 10-minute video plus a WhatsApp cheat card for teachers. 1-hour live session for Ops. Ops "champions" run floor support. Gate: auto-rate ≥ 65%, SLA breach < 10%. |

### D4. Risks

- **Wrongly auto-approving a bad case.** The policy is conservative: any doubt means a human. Money and identity are never decided by the LLM.
- **Teachers resist change.** They lose a chore, not gain one. The WhatsApp-only flow and an orphan scorecard tied to the existing coaching process help.
- **Over-messaging parents.** Cap at one proactive message per event and quiet hours (21:00-08:00 local, GCC/US aware).
- **Sample is tiny.** All rates here are directional. Phase 0 re-baselines them on the full table before any target is locked.
- **Data assumptions.** The engine assumes `active_students_count` includes the renewing child. Phase 0 must confirm this with engineering before the false-"full" fix goes live.

Repo: `src/reconcile.py` (engine) · `prompts/` (triage agent, policy) · `output/` (results) · `data/` (5 source CSVs).